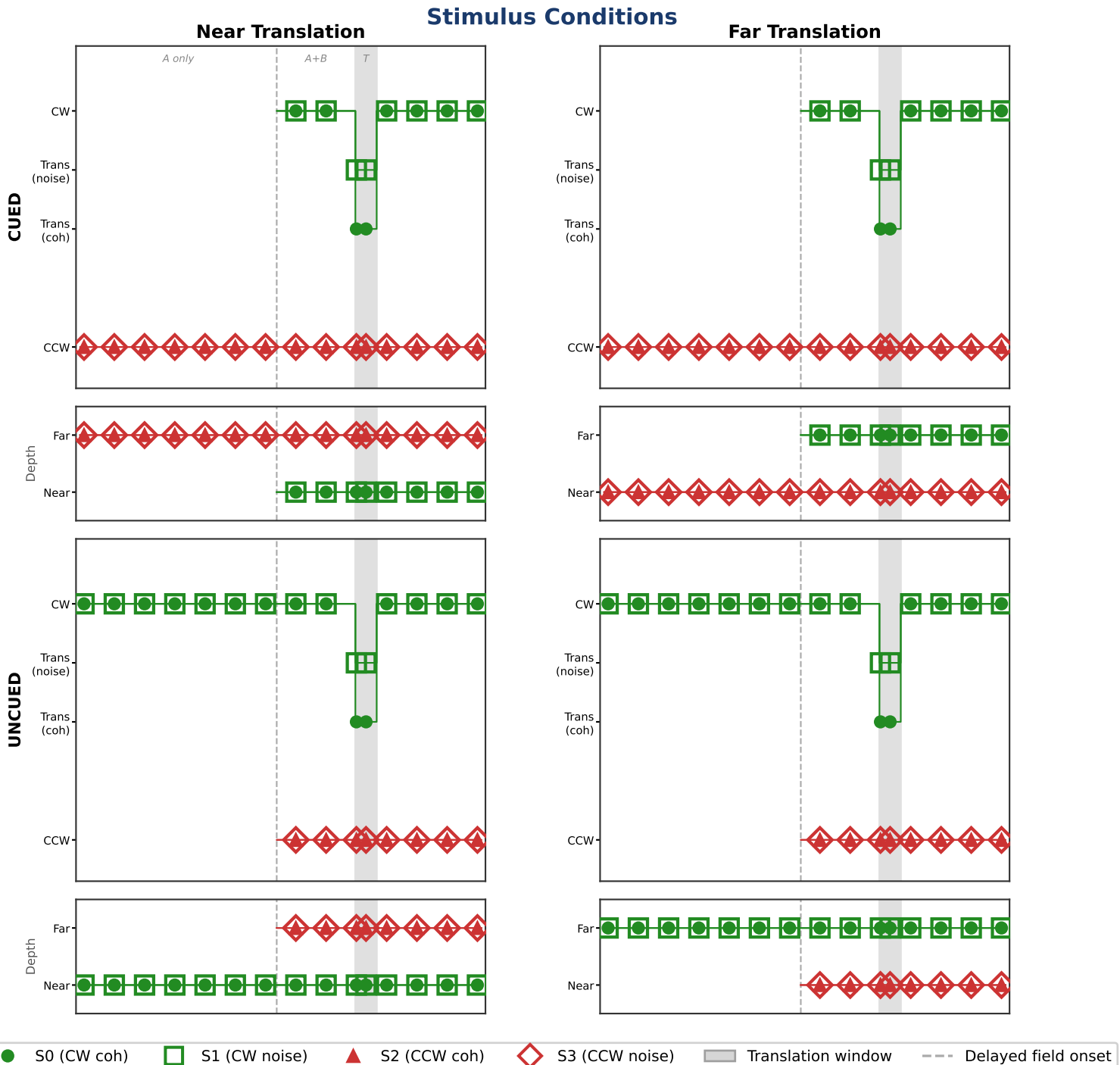


VRDots — DepthParam (Parametric Depth, No Swap)

260402_0716 (0.03 m) • 260402_0757 (0.05 m) • 260402_0624 (0.10 m) • 260402_0656 (0.15 m) — R/G balanced, no swap, n=128/session

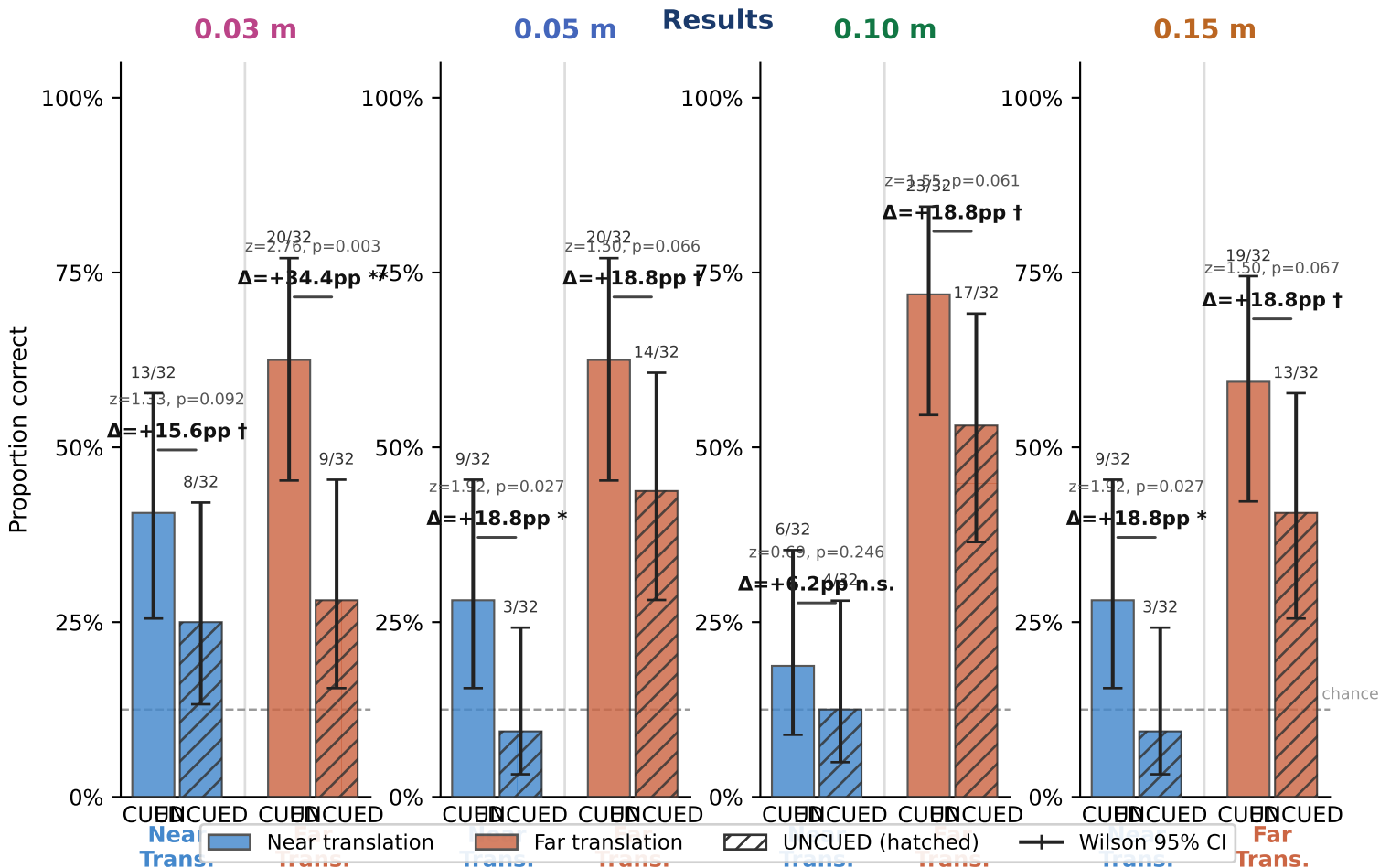
The DepthParam experiment measured dot cueing across four depth separations (0.03, 0.05, 0.10, 0.15 m) to characterize the Near/Far asymmetry as a function of stereoscopic depth magnitude. Unlike DepthSwapCtrl, no depth swaps were applied (SwapType = N only). Fields were two-color and rotation-balanced (DelayedFieldColor R/G, RotCfg 0/1 crossed), yielding 128 trials per session, approximately 32 trials per Cond × DelayedFieldDepth cell. All four sessions were collected in a single morning (260402). Depth separation is measured as the full distance between the two planes (each field is $\pm$ half the separation from the fixation depth at 2.0 m).

Trajectory diagrams show one representative pairing: S0●/S1□ = initially CW field (green); S2▲/S3◇ = initially CCW field (red). Columns: Near translation (translating field at Near) | Far translation (translating field at Far). Motion tracks (upper) and depth tracks (lower) both use the 4-subfield marker convention. In the depth track, S0/S1 and S2/S3 share the same depth level so their symbols are superimposed — the open larger symbol (S1□ or S3◇) frames the filled smaller one (S0● or S2▲).



VRDots — DepthParam: Results

Analysis: Near/Far labels use the translating-field-depth convention. CUED Near = delayed field at Near, translating Near. UNCUEd Near = non-delayed field at Near, translating Near (delayed field at Far). Proportion correct pooled across color, rotation, and direction (≈ 32 trials/cell per session). One-tailed z-test: CUED > UNCUEd.



Findings

Overall cueing was present in all four sessions (range +12.5 to +25.0 pp). The 0.03 m session showed the strongest overall effect (+25.0 pp, $p=.002$), with significant Far cueing (+34.4 pp, $p=.003$) and marginal Near cueing (+15.6 pp, $p=.094$). At 0.05 m: overall +18.8 pp ($p=.022$), with Near (+18.8 pp, $p=.025$) and Far (+18.8 pp, $p=.067$) both contributing. At 0.10 m: overall +12.5 pp ($p=.063$), with Near CUED performance near floor (18.8%) making Near cueing undetectable (+6.2 pp, n.s.) while Far cueing was marginal (+18.8 pp, $p=.063$). At 0.15 m: overall +18.8 pp ($p=.017$), with Near cueing significant (+18.8 pp, $p=.025$) and Far marginal (+18.8 pp, $p=.067$). The Near/Far accuracy asymmetry (Near task harder) is visible at all depths but is most severe at 0.10 m where Near CUED drops to 18.8% (near chance). The pattern across depths is not monotonic: 0.03 m produces the highest Far cueing (+34.4 pp) despite having the smallest disparity, suggesting that very large depth separations may degrade Near-plane legibility without proportionally increasing Far-plane cueing.

Notes

Notes: (1) $n=32$ /cell is low; all effects should be interpreted cautiously. Second sessions are planned at each depth. (2) Unlike DepthSwapCtrl, these sessions are two-color (R/G balanced), making direct magnitude comparisons with DepthSwapCtrl confounded by color-axis differences. (3) The 0.05 m DepthParam session (260402_0757) provides a same-depth/different-color comparison with DepthSwapCtrl (also 0.05 m). DepthParam 0.05 m overall cueing: +18.8 pp* vs. DepthSwapCtrl bino: +12.5 pp***, consistent with two-color stimuli yielding stronger cueing than same-color. (4) Three aborted sessions (260402_0549, 0610, 0618) preceded the complete sessions and are excluded.